

Mental Causation

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What's at stake with mental causation? Why should we care about vindicating it?

1. The possibility of human agency, and hence the appropriate ascription of moral responsibility, requires that our mental states have causal effects in the physical world. For voluntary actions to even be possible, our beliefs and desires must somehow cause our limbs to move in appropriate ways, thereby causing the objects around us to be rearranged.
2. The possibility of human knowledge presupposes the reality of mental causation: perception requires the causation of perceptual experiences and beliefs by objects and events around us; reasoning involves the causation of new belief by old belief; and memory is a causal process involving experiences, physical storage of the information contained therein, and its retrieval. If you take away perception, memory, and reasoning, you pretty much take away all of human knowledge.

So while determinism threatens moral responsibility and skepticism puts human knowledge in peril, the stakes are higher with mental causation, for this problem threatens to take away both agency and cognition in one swoop.

Kim's Causal Exclusion Argument

The McLaughlin paper starts with him saying, “Kim (1998b) has defended a bold position concerning mental causation: the nature of our world is such that every mental event is either a physical event or an epiphenomenon.” This is, of course, referring to his causal exclusion argument. So let's begin by exploring that. Below is a more spelled out version of the argument Kim presented in V in his (1989):

First, the core commitments of the non-reductive physicalist:

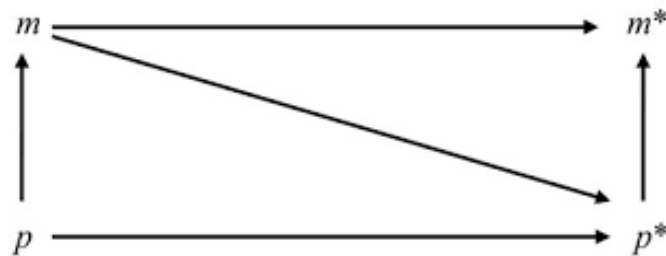
1. Mind-body supervenience: If any system s instantiates a mental state M at t , there necessarily exists a physical state P such that s instantiates P at t , and necessarily anything instantiating P at any time instantiates M at that time
2. Physical causal closure: If a physical event e has a cause at time t , then e has a physical cause at t .
3. Causal efficacy: Mental properties have causal efficacy—that is, their instantiations can and do cause other properties, both mental and physical, to be instantiated
4. Irreducibility: Mental properties are not reducible to, and are not identical with, physical properties

And one more general principle:

Exclusion: No single event can have more than one sufficient cause occurring at any given time—unless it is a genuine case of causal overdetermination

With this in mind, I take the causal exclusion argument to look something like this:

1. Psychological causation occurs, expressed as $M \rightarrow M^*$ (Causal efficacy)¹
2. For mental property M^* , there exists a physical property P^* as its supervenient base (Mind-body supervenience)
3. If $M \rightarrow M^*$, then $M \rightarrow P^*$ (1 & 2)²
4. P^* must have a physical cause occur at the time M occurs (Physical causal closure)
5. For mental property M , there exists a physical property P (Mind-body supervenience)
6. If $M \rightarrow P^*$, then $P \rightarrow P^*$ (3, 4, 5)
7. If $M \rightarrow M^*$, then $M \rightarrow P^*$ and $P \rightarrow P^*$ (1, 3, 6)
8. $M \neq P$ (Irreducibility)
9. P^* has two distinct, independently sufficient causes (7 & 8)
10. Assuming P^* is not a case of causal overdetermination, we must eliminate either M or P as a cause of P^* (Exclusion)
11. $\neg(M \rightarrow P^*)$ (Physical causal closure)
12. If $\neg(M \rightarrow P^*)$, then $\neg(M \rightarrow M^*)$ (3)
13. If $M \rightarrow M^*$, then $\neg(M \rightarrow M^*)$ (1, 7, 11, 12)



So either $M = P$, or $\neg(M \rightarrow M^*)$; mental events are physical events or they are epiphenomenon. Thus we reach Kim's bold position: "If nonreductive physicalists accept the causal closure of the physical domain, therefore, they have no visible way of accounting for the possibility of psychophysical causation. This means that they must either give up their antireductionism or else reject the possibility of psychophysical causal relations." (p.47)

¹ Note that since properties do not enter into causal relations, what is meant by the claim that M causes M^* is that the instantiation of some event with mental property M causes the instantiation of some other event with mental property M^* .

² M must cause P^* if we want $M \rightarrow M^*$. Since P^* is both necessary and sufficient for M^* , then the instantiation of P^* would have also brought upon the instantiation of M^* even if M had not occurred at a time before M^* was instantiated. Thus, if supervenience holds, in a counterfactual world in which P^* holds but M does not occur, then M^* would have been instantiated regardless. This seems to render mental properties as causally irrelevant to the instantiation of other mental properties. The only way to vindicate causal efficacy is to say that the instantiation of M caused the instantiation of P^*

This argument crucially informs the insights from the McLaughlin paper, which builds upon its conclusions. However, briefly, note that in the paper, Kim, in trying to deny the causal overdetermination option, writes:

“Moreover, the overdetermination idea seems to violate the causal closure principle as well: in the counterfactual situation in which the physical cause does not occur, the closure principle is violated. For the idea that the mental and the physical case are each an independent sufficient cause involves the acceptance of the counterfactual that if the physical case had not occurred, the mental cause would have occurred and caused the physical effect. This is in violation of the causal closure principle.” (p. 44)

Problem: it seems to me that if mind-body supervenience holds, there can be no possible world in which M can occur without its supervenient base, P, occurring as well, i.e. if we take away the supervenient base of M, which is P, then M would not have occurred. So without P, we cannot have $M \rightarrow P^*$, thus not violating physical causal closure.

Perhaps then overdetermination is an option for the non-reductive physicalist, but for sake of discussion, let's just assume it isn't.

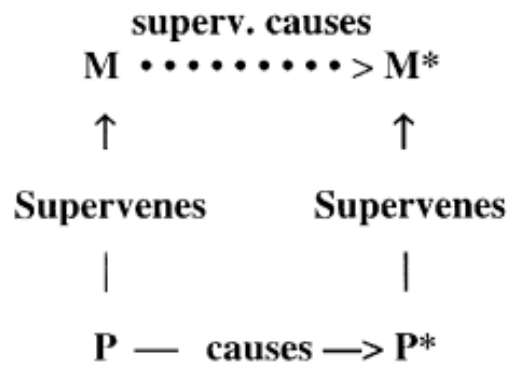
Does Mental Causation Require Psychophysical Identities?

What would it take for the nonreductive physicalist (NRP) to provide an account of mental causation?

- 1) Provide an account of realization that says every mental event is realized by a physical event without entailing mental events are physical events.
- 2) Explain how the mental events realized by physical events can have genuine causal power in virtue of their mentality in spite of physical causal closure (formulated above) and the physical effects principle (if a mental event is a cause of another mental event, then it is a cause, or is among the causes, of some physical event or other).

McLaughlin thinks there is currently no formulation of NRP that meets both of these conditions, and aims to show so in this paper. But first, a remark on a core assumption made by the NRP theorist that isn't challenged by McLaughlin:

“NRP theorists maintain that whenever a mental event is a cause or effect of another event, its physical realizer is, too. Because when a mental event is a cause of another event its physical realizer is as well, there is always a kind of causal overdetermination involved in mental causation. But NRP theorists claim that the kind in question is different from run-of-the-mill overdetermination, the kind involved when, for instance, two fatal blows cause a death. In run-of-the-mill overdetermination, the causal chain from one overdeterminer to the effect is independent of the causal chain from the other overdeterminer to the effect. But mental-to-mental and mental-to-physical causal transactions are always dependent in a certain way on entirely physical causal transactions involving physical realizers of the mental events. Causal transactions involving mental events are implemented by causal transactions involving their physical realizers.” (p. 69)



So I take these sorts of causal processes to look something like the above figure (although the relation between P and M could be strengthened so that P realizes M instead of merely saying M supervenes on P). Remember, if we endorse physical causal closure and mind-body supervenience, then we are led to saying that if $M \rightarrow M^*$, then $M \rightarrow P^*$ and $P \rightarrow P^*$. In response, the NRP theorist could say that, while M doesn't have the same causal status of P in relation to P^* , it is nevertheless a "derivative" cause of P^* *in virtue of being realized by P*. Does accepting interlevel causal relations vindicate the causal efficacy of mentality, as the NRP theorist wants to say it does, i.e. does supervenient causation exist? I suspect the answer to be no, but I don't have a definite argument for my intuition. I will, however, offer some reasons for why I think the existence of such causation is dubious.

1. If we accept the causal relations in the above figure, we can ask, "what caused the instantiation of M^* at time t_1 ? Was it because its physical base, P^* , was instantiated at t_1 , or was it because of M superveniently causing it at time t_0 ? It seems to me that as long as the physical base P^* was instantiated at t_1 , M^* will also be instantiated regardless of what happened before t_1 . Additionally, unless P^* (or some other physical base) were present at t_1 , M^* cannot be present. So, if P^* being instantiated at t_1 is fully sufficient for M^* at t_1 , then it seems to me that anything happening before t_1 will be irrelevant to some event having M^* , unless it were to instantiate P^* .
2. But were M to instantiate P^* , we return to the causal exclusion problem. Adding an arrow between M and M^* and saying that M superveniently causes M^* does nothing but mislead us into potentially thinking that M does some real causal work.

Functional Realization and Mental Causation

McLaughlin's main point, as I take it, is that if functional properties really exist and are higher order properties of its core realizers, then how can it contribute any causal work that isn't already done by its core realizers if the functional properties are not identical to the core realizers? This was quite a long paper and a lot of topics were touched on, many of which I'm not too familiar with, so I will divert my focus to a few passages I found interesting.

1. “On the sort of view described earlier, events participate in causal interactions in virtue of their constitutive properties. And if functional properties are not constitutive properties of events, but only characterizing properties of events, then events do not participate in causal relations in virtue of being functional events.” (p. 80)

It seems to me that at least sometimes characterizing properties of events can figure in causal explanation, but only if they are reducible to or are identical with some core realizer. Take, for example, the temperature of a gas. I take that temperature is a functional property, defined by, among other things, how a system interacts with a thermometer, how it behaves in heat exchange, its relation to pressure or volume, etc. It can be realized by several different microstates, but temperature does not describe the microstate of the system it characterizes. So say that the property of being 70 degrees is having the property of having some physical property that plays a certain causal role when exposed to a thermometer. It seems to me that this does not disqualify temperature from entering into causal explanations. Say that the pressure in a sealed container is increasing; in giving a causal explanation of this phenomena, surely we want to cite the fact that the temperature of the gas within the container is increasing. It seems weird to say that temperature was causally irrelevant to the increasing pressure in the container since it is not a constitutive property of the system. This isn't a response on behalf of the role-functionalist, since they deny that functional mental properties can be reduced to or are identical with their physical core realizers, but I thought it was too strong to say only constitutive properties of events participate in causal relations.

2. “Thus, suppose that the Humean view is indeed wrong, and that some structures are essentially killer structures, and so essentially endow the power to kill. The question would still remain why the state of having some or other killer structure is a state that is itself a killer state – a state that endows the power to kill – given that it is not identical with any of the killer structures. Why think that the property of having some property or other that essentially endows the power to kill is itself a property that endows the power to kill? Similarly, even if all the bases or core realizers of a functional property, water solubility, essentially endow the (conditional) power to cause dissolvings in water, what reason is there to think that the functional property does – that water solubility itself does?” (p. 86)

A potential response on behalf of the NRP theorist: let's stipulate, for some core realizer of F, there is some metaphysically possible world in which the laws of nature were different and so the core realizer doesn't realize F and thus doesn't cause some effect E. In this example, however, the microphysical state of the core realizer is held constant so that, besides not realizing F, it is identical to the core realizer of F in the actual world. If we think of causation as a matter of counterfactual dependency, then it is the case that without F being realized, E would not have occurred, thus giving F some genuine causal power.

I worry however, that this response depends on a form of separatism that we have reasons to reject, so I'm not confident that it'll work for the NRP theorist.

3. “The reason is that if S is a sufficient set for E, then some proper subset of S will be too, namely, the set that includes all of the members of S except F. That set will include the realizer of F, and so will suffice for E. Thus, functional events cannot figure in minimal sufficient sets.” (p. 99)

This seems to be an unacceptable form of separatism invoked by McLaughlin. In all nomologically possible worlds, if the realizer of F is instantiated, then F will be as well. And if we think the realizer of F , which I will call F_r , is not metaphysically necessary for F because there could be some possible world in which the laws of nature were different so that F_r could not play the relevant role to realize F , in such a world, why should we think F_r can cause E ? Since F_r cannot realize F in that possible world, then clearly F_r plays a different causal role in that world. And if F_r plays a different causal role, it's not clear to me why we should think E will obtain in that world as a result of being caused by F_r .

A Final Worry

Can psychophysical identities save mental causation? It seems intuitive, especially given Kim's exclusion argument, but I think there is some reason to think it can't if causation is hyperintensional.

Example: suppose John were to (legally) purchase alcohol in the U.S. In giving a causal explanation of this event, it seems the fact that John is 21 is causally relevant. However, it does not seem causally relevant that John's age was the product of 3 and 7, even though that necessarily refers to the same thing as 21. It seems to me that being the product of prime numbers or something of that sort is just not causally relevant to the story of why John was able to purchase alcohol. It was because John was 21, not because his age was the product of the 2nd and 4th prime numbers, even if they are identical. Perhaps this intuition is mistaken, but if it is not, we can now generalize to the case of mental causation.

Suppose I wince and cry out at some time after touching a hot stove. In giving a causal explanation of this event, you can cite the fact that I was in pain when I touched the hot stove. However, if causation is hyperintensional, it's at least possible that the fact that, say, my C-fibers firing is causally irrelevant to an explanation as to why I winced and cried out later, even if C-fiber stimulation and pain are identical. So even if $M = P$, and $P \rightarrow P^*$, it is not guaranteed that $M \rightarrow P^*$.

So there's at least a challenge for the reductive physical to either deny that causation is hyperintensional or to say that it does not apply to the case of mental causation. I'm not at all familiar with these discussions, so I leave this as an open question of whether it works or not.